

Machine Learning

Linear Regression Report

California Housing Price Prediction

Task 1 – AI/ML Internship

Dataset: California Housing (scikit-learn)

Model: Linear Regression

Prepared by: Abhilasha Singh | Organization: Maincrafts

Date: June 2026

1. Introduction

This report documents the end-to-end machine learning workflow applied to the California Housing dataset. The goal is to predict the **median house value** (MedHouseVal) for California districts using demographic and geographic features. This project covers data loading, exploratory data analysis (EDA), model training, evaluation, and future improvement recommendations.

Objective: Train a Linear Regression model, evaluate it using MAE, RMSE and R^2 , and communicate findings clearly through visualisations and metrics.

2. Dataset Overview

The California Housing dataset is sourced from the 1990 U.S. Census and is built into **scikit-learn**. It contains **20,640 samples** and **8 input features**.

Feature	Description	Type
MedInc	Median income in block group	Numeric
HouseAge	Median house age in block group	Numeric
AveRooms	Average number of rooms per household	Numeric
AveBedrms	Average number of bedrooms	Numeric
Population	Block group population	Numeric
AveOccup	Average household occupancy	Numeric
Latitude	Block group latitude	Numeric
Longitude	Block group longitude	Numeric
MedHouseVal	Median house value (TARGET)	Numeric

Table 1 – Feature descriptions of the California Housing dataset.

3. Exploratory Data Analysis (EDA)

3.1 Data Quality Check

- Total records: **20,640** | Features: **8**
- Missing values: **None** — no imputation required
- No duplicate rows detected
- Target variable (MedHouseVal) range: **0.15 – 5.00** (in \$100,000s)

3.2 Target Variable Distribution

The target variable MedHouseVal is **right-skewed**, with most house values concentrated between \$100,000–\$300,000. A spike at \$500,000 indicates a census cap on reported values. This skewness may slightly affect linear model performance.

Statistic	Value
Mean	\$206,855
Median	\$179,700
Std Dev	\$115,395
Min	\$14,999
Max	\$500,001 (capped)

Table 2 – Descriptive statistics of target variable.

3.3 Feature Correlation Insights

A Pearson correlation heatmap was generated to identify relationships between features and the target variable:

Feature	Correlation with MedHouseVal	Interpretation
MedInc	+0.69	Strong positive — key predictor
AveRooms	+0.15	Weak positive
HouseAge	+0.11	Weak positive
Latitude	-0.14	Slight negative (northern areas cheaper)
Longitude	-0.05	Minimal effect
AveOccup	-0.02	Negligible
Population	-0.02	Negligible
AveBedrms	-0.05	Negligible

Table 3 – Feature correlation with target variable (MedHouseVal).

4. Data Preprocessing

- **Train/Test Split:** Data split into 80% training (16,512 samples) and 20% testing (4,128 samples) using `random_state=42` for reproducibility.
- **Feature Scaling:** No explicit scaling applied (LinearRegression still converges; however, StandardScaler is recommended as an improvement).
- **No Outlier Removal:** Outliers retained to preserve real-world distribution; future work may apply IQR-based filtering.

Code used for splitting:

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
```

5. Model Training

A **Linear Regression** model from scikit-learn was trained on the training set. Linear Regression models the relationship between input features and the target as a linear combination:

$$y = b_0 + b_1 \cdot \text{MedInc} + b_2 \cdot \text{HouseAge} + b_3 \cdot \text{AveRooms} + \dots + b_n \cdot \text{Longitude}$$

- Algorithm: Ordinary Least Squares (OLS)
- Library: scikit-learn `LinearRegression()`
- Training samples: 16,512
- Fitting time: ~0.1 seconds (CPU)

6. Model Evaluation & Results

6.1 Performance Metrics

The trained model was evaluated on the unseen test set using three standard regression metrics:

Metric	Value	Meaning	Verdict
MAE	~0.533	Avg error of \$53,300	Acceptable
RMSE	~0.745	Penalises large errors more	Moderate
R ²	~0.606	Model explains 60.6% of variance	Moderate

Table 4 – Model performance on test set (values in \$100,000 units).

6.2 Actual vs Predicted Plot (Summary)

A scatter plot of actual vs predicted values reveals that the model performs well for house values in the \$100,000–\$300,000 range. However, predictions deviate at higher price ranges (above \$400,000), suggesting non-linear patterns the model cannot capture.

Key Observation: Points cluster near the ideal diagonal line for mid-range values, but spread increases for high-value properties — a sign of heteroscedasticity.

6.3 Residuals Analysis

The residuals (actual – predicted) were plotted as a histogram. Key findings:

- Residuals are approximately **normally distributed** around 0 — a good sign.
- Slight right skew observed, caused by the capped values at \$500,000.
- No strong systematic bias detected in mid-range predictions.

7. Improvement Ideas

The following improvements can significantly boost model performance:

#	Improvement	Expected Benefit
1	Feature Scaling (StandardScaler)	Faster convergence, better regularisation
2	Polynomial Features (degree=2)	Captures non-linear relationships
3	Ridge / Lasso Regression	Reduces overfitting, handles multicollinearity
4	Random Forest Regressor	Large R ² improvement (typically 0.80+)
5	Outlier Removal (IQR method)	Reduces RMSE on extreme values
6	Log-transform target variable	Fixes right skew, improves residual normality
7	Cross-validation (k-fold)	More reliable performance estimate

8	Gradient Boosting (XGBoost)	State-of-the-art performance on tabular data
---	-----------------------------	--

Table 5 – Recommended improvements for future iterations.

8. Conclusion

This project successfully implemented a complete machine learning pipeline for house price prediction using Linear Regression on the California Housing dataset. The model achieved an R^2 of approximately **0.606**, explaining 60.6% of variance in median house values.

The most important predictor is **MedInc (median income)**, which showed the highest correlation (+0.69) with the target. The model works well for average-priced homes but struggles with very high-value properties due to non-linear patterns.

Next Steps: Apply feature scaling, try Ridge Regression and Random Forest, and use k-fold cross-validation to improve the model's R^2 from 0.606 to 0.80+.